ZentroMart VoiceOps Support Agent

VoiceOps 2026 capstone report | 25 October 2026

Project summary

This project builds a policy-grounded customer support agent for the fictional ZentroMart retailer. It answers questions from company documents, uses local catalog and delivery data for exact facts, verifies callers before disclosing order details, and can process voice messages with Gnani speech services.

The text model is Gemini 3.5 Flash Lite. Retrieval uses Gemini embeddings with a lexical fallback. The voice path uses Gnani STT and TTS only. The demo language scope is English, Hindi, and Hinglish.

Implementation

Step	Implementation
1. Prompt baseline	A concise support prompt was evaluated on the 52 practice questions before adding retrieval.
2. Grounding	Markdown sections retain their evidence IDs and policy metadata. Hybrid retrieval ranks policy evidence, and structured product or pincode facts come from local tables. Unsupported questions are handled without inventing policy.
3. Conversation	The agent tracks the issue and order context, verifies before sharing private order facts, allows one verification retry, handles topic changes, and creates a ticket for handovers.
4. Voice	The batch runner sends 16 kHz mono audio to Gnani STT, passes the transcript through the same support agent, and returns Gnani TTS audio with text and evidence records.

The safety boundary is intentionally narrow: the supplied mock order tool is used only for synthetic capstone data; order information is disclosed only after the tool verifies the caller; verification values are redacted from saved logs; and a handover is reported only after ticket creation succeeds. Gnani is the only speech provider.

Evaluation results

Evaluation	Approximate pass rate	Fact coverage	Other result
Step 1 prompt baseline, 52 questions	30.8%	24.1%	No retrieval evidence
Step 2 RAG, 52 questions	82.7%	89.7%	92.3% decision accuracy; 100% evidence overlap in the local scorer
Step 4 ASR-style text v2, 52 questions	92.3%	91.4%	98.1% decision accuracy; 100% evidence overlap in the local scorer

The RAG run improved the approximate pass rate by 51.9 percentage points over the prompt baseline. Retrieval evaluation on 47 evidence-bearing questions recorded 47/47 Hit@5, MRR@5 of 0.952, and evidence recall@5 of 0.805. The full 23-conversation practice set passed 96/96 turns in the local string and routing checks. A later targeted tool-argument check passed all 4 turns of one failed-verification conversation.

These are local practice metrics, not scores from the organizer's hidden set. The scorer checks listed facts, decisions, and evidence overlap approximately; it cannot establish that every answer is semantically correct. The ASR-style run uses the dataset's text transcripts and is not an ASR accuracy benchmark.

What changed during development

Free-tier Gemini request limits and temporary 503/504 service errors interrupted early batch runs. The RAG question runner was made resumable, so completed responses can be reused. Embedding quota and request-parameter problems initially triggered lexical retrieval; the final hybrid retrieval run completed and found the expected evidence in all 47 evidence-bearing questions.

The first Hinglish TTS setting used hi-en, which the installed Gnani SDK rejected. The batch completed STT and answer generation but failed during TTS. The Hinglish configuration was corrected to use auto, which the SDK accepts for mixed-language synthesis. English, Hindi, and Hinglish voice samples have since produced reply audio.

Limitations and next work

The final demo still needs to be recorded. The current batch command applies one language pair to all files in an invocation, so a mixed-language folder must be split into separate batches. Voice checks cover a small number of self-recorded samples; they do not establish quality across accents, devices, or noisy conditions. Further semantic review of low-scoring practice answers and testing with fresh audio would improve confidence.

Before submission, run the documented setup from a clean environment, confirm the organizer-facing audio-folder command and output structure, review the final report, record a demo of at most two minutes showing a grounded answer with a follow-up and a handover or an "I don't know" response, and package the code, README, .env.example, dependencies, report, demo, and sample outputs. Exclude .env, API keys, the dataset copy, virtual environments, and personal recordings.